

Night Observations from base - no optimization

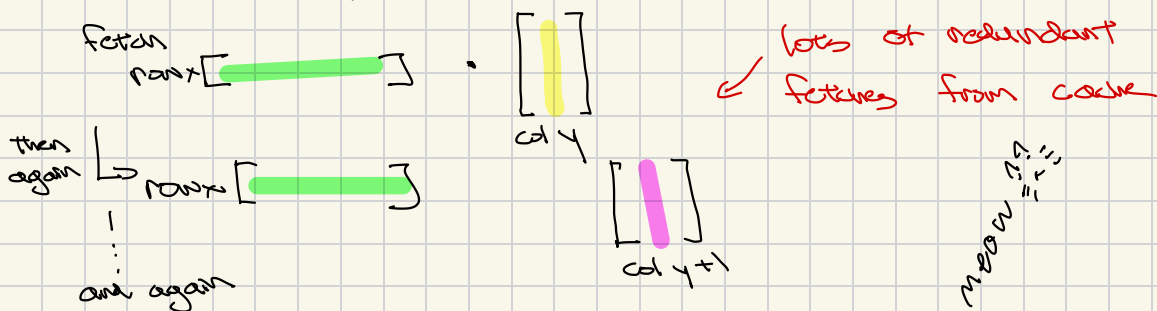
- Testing using 512×512 matrix mult

Finding

- 94% Achieved Occupancy
 - Good, GPU busy
- 80% Mem Throughput vs 80% Compute Throughput
 - Good, balanced, neither is bottlenecking other
- 1% Dram throughput, 81% L1 throughput, 68% L2 throughput
 - Good, data is mostly stored in cache vs ram
 - ↳ better for memory locality
- Branch Efficiency 100%, Avg Divergent Branches: 0
 - Good, no warp divergence

Issues

- Scheduler Statistics: 61% no eligible
 - Threads only issuing instructions only 38% of cycles
 - Waiting on memory to come back*
- Problem
 - A lot of redundant fetching as each col/row is pulled multiple times



- Uncoalesced memory access 13.98% est speedup
 - Comes from $\text{dev.mem}[\text{ml_row} * k + i]$

lot of mem addresses away

Next Steps: 1. Shared memory and tiling

Some more benchmarking

Tested base kernel vs cuBLAS (Nvidia Standard)

Tried 32, 64, 128, 256, 512, 1024 num of threads launched across 512×512 , 2048×2048 , 4096×4096 matrices

Observations

- Kernel actually beats cuBLAS at 512×512
 - Prob just more lightweight kernel
 - cuBLAS initialization overhead
 - Had to warm up first
- Execution time around 50% of cuBLAS at 2048×2048
25-30% of cuBLAS at 4096×4096
- Thread count doesn't really affect execution time
 - Best performance: 64 threads at 512×512
256 threads at 2048×2048
512 threads at 4096×4096

↑
This is prob due to being memory bound over compute bound

Scheduler can schedule more threads but they're all waiting on memory as seen from the insight

Solution: Reduce access to global memory

Roofline Analysis

- 4096 x 4096 at 512 threads

$$\text{One element} = \sum_{k=0}^{4096} A[i, k] \times B[k, j]$$

(i, j)

so 4096×2 FLOPs per output elem
 | addition
 | multiplication

$$\text{FLOPs} = \underbrace{4096 \times 4096}_{\substack{\uparrow \\ \text{Total \# of output} \\ \text{elements}}} \times \underbrace{2 \times 4096}_{\substack{\text{FLOPs per output} \\ \text{element}}} = 137.44 \text{ GFLOPs}$$

Bytes moved: 7.47 GB + 944.52 MB = 8.4 GB
 - Read - Write

Time taken: 81.02 ms - from ncu
 total bytes moved

Compute Throughput: $\frac{137.44 \text{ GFLOPs}}{81.02 \text{ ms}} = 1.696 \text{ TFLOPs/s}$

Compute Time Floor: $\frac{137.44 \text{ GFLOPs}}{56 \text{ TFLOPs/s}} = 2.45 \text{ ms}$
 max of 5680

DRAM

Memory Throughput: $\frac{8.4 \text{ GB}}{81.02 \text{ ms}} = 103.8 \text{ GB/s}$

DRAM

Memory Time Floor: $\frac{8.4 \text{ GB}}{960 \text{ GB/s}} = 8.75 \text{ ms}$

L2 Time Floor: $\frac{276.25 \text{ GB}}{3.4 \text{ TB/s} / 0.94} = 76.3 \text{ ms}$

Actual BW percentage of peak BW

$\frac{276.25 \text{ GB}}{3.4 \text{ TB/s}} = 81.0 \text{ ms}$

very close to actual

Confirmed L2 is bottleneck